

BCA 2nd Semester Exam., 2014

PROBLEM SOLVING TECHNIQUE AND
PROGRAMMING IN C

Time : 3 hours akubihar.com Full Marks : 60

Instructions : akubihar.com

- (i) **All** questions carry equal marks.
- (ii) There are **SEVEN** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question Nos. 1 and 2 are compulsory.

1. Select the appropriate option from the multiple choices given below (any six) : akubihar.com

- (a) The C language has been developed by
 - (i) Patrick Nughton
 - (ii) Dennis Ritchie
 - (iii) Ken Thompson
 - (iv) Martin Richards
- (b) The C programs are converted into machine language using
 - (i) an assembler
 - (ii) a compiler
 - (iii) an interpreter
 - (iv) an operating system

- (c) The declaration of C variable can be done
 - (i) anywhere in the program
 - (ii) in declaration part
 - (iii) in executable part
 - (iv) at the end of program
- (d) The range of character data type is
 - (i) -128 to 127
 - (ii) 0 to 255
 - (iii) 0 to 32767
 - (iv) -32767 to 32767
- (e) What is the remainder of 8% 10?
 - (i) 8
 - (ii) 2
 - (iii) 1
 - (iv) 0
- (f) What is the ASCII range for A to Z letters?
 - (i) 65 to 90
 - (ii) 48 to 57
 - (iii) 97 to 122
 - (iv) None of the above
- (g) Each case statement in switch() is separated by
 - (i) break
 - (ii) continue
 - (iii) exit()
 - (iv) goto

- (h) A character array always ends with
- (i) null character akubihar.com
 - (ii) question mark
 - (iii) full stop
 - (iv) ampersand symbol
- (i) Recursion is a process in which a function calls
- (i) itself
 - (ii) another function
 - (iii) main() function
 - (iv) None of the above
- (j) The member variable of structure is accessed by using akubihar.com
- (i) dot operator
 - (ii) arrow operator
 - (iii) asterisk operator
 - (iv) ampersand operator

2. Answer any *three* of the following questions :

- (a) What are pointers? Why are they important?
- (b) What is recursion? Explain its advantages.
- (c) Distinguish between Break and Continue.
- (d) What are the uses of strcpy() and strcmp() functions?
- (e) What are the relational operators?

- 3. Write a program to enter a four-digit number. Display the digits of the number in reverse order.
- 4. Write a program to find the smallest number in an array of 10 elements. akubihar.com
- 5. Write a program to find length of a given string using pointer.
- 6. Write a program to check whether a given number is Armstrong or not.
- 7. Write a program to print table of a given number.

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BCA 2nd SEMESTER EXAM., 2015
PROBLEM SOLVING TECHNIQUE & PROGRAMMING IN C CODE- 303204

Time: 3 hours

Full Marks: 60

Instructions:

- i. The Marks are indicated in the right -hand margin.
- ii. There are **SEVEN** questions in this paper.
- iii. Attempts **FIVE** question in all.
- iv. Question Nos. **1** and **2** are compulsory.

Group - A

1. Answer any six of the following :

2*6=12

(a) Which of the following special symbols allowed in a variable name

- | | |
|------------------|---------------------|
| (I) * (asterisk) | (II) (pipeline) |
| (III) - (hyphen) | (IV) _ (underscore) |

(b) Is there any difference between following declarations?

(1) extern int fun(); **(2) int fun();**

- (I) Both are identical
- (II) No difference, except **extern int fun();** is probably in another file
- (III) **int fun();** is overridden with **extern int fun();**
- (IV) None of the above

(c) When we mention the prototype of a function?

- | | |
|-------------------|----------------|
| (I) Defining | (II) Declaring |
| (III) Prototyping | (IV) Calling |

(d) In mathematics and computer programming, which is the correct order of mathematical operators?

- (I) Addition, subtraction, multiplication, division
- (II) Division, multiplication, addition, subtraction
- (III) Multiplication, addition, division, subtraction
- (IV) Addition, division, modulus, subtraction

(e) Which of the following cannot be checked in a switch-case statement?

(I) Character

(II) Integer

(III) FLoat

(IV) Enum

(f) Which of the following is the correct usage of conditional operators used in C?

(I) $a > b ? c = 30 : c = 40;$

(II) $a > b ? c = 30;$

(III) $\max = a > b ? a > c ? a : c : b > c ? b : c$

(IV) $\text{return } (a > b) ? (a : b)$

(g) Which of the following are unary operators in C?

1. $!$

2. *Sizeof*

3. $\sim$

4. $\&\&$

(I) 1, 2

(II) 1, 3

(III) 2, 4

(IV) 1, 2, 3

(h) Input/output function prototypes and macros are defined in which header file?

(I) conio.h

(II) stdlib.h

(III) stdio.h

(IV) dos.h

(i) What is stderr?

(I) Standard error

(II) Standard error types

(III) Standard error streams

(IV) Standard error definitions

(j) What is the purpose of fflush() function?

(I) Flushes all streams and specified streams

(II) Flushes only specified streams

(III) Flushes input/output buffer

(IV) Flushes file buffer

Group - B

2. Answer any three of the following:

4*3=12

(a) What do you mean by function? How we can use the user defined functions?

akubihar.com (b) What is a storage class? Discuss the various storage classes.

- (c) What is an array? What is 2-dimensional array? How 2-dimensional arrays are initialized? Also explain how array elements are accessed in a 2-dimensional array? give explanation with example.
- (d) What do you mean by the user defined data types? Give some differences between structure and union.
- (e) Write the difference between functions and macros.

Group - C

Answer any three of the following:

12*3=36

- 3.** How we open and close an existing file in C? Also discuss the various modes to open a file.
- 4.** Write a program to check whether the number is prime or Armstrong with the help of user defined functions.
- 5.** Give the advantage of array using pointers. Write a program to access element of an array by using pointers.
- 6.** What is an operator? What are different types of operators? Discuss the various operators available in C
- 7.** What do you mean by a recursion? Write a program to print a Fibonacci series using recursion up to 1000 numbers.

*** * ***

BCA 2nd Semester Exam., 2016

PROBLEM SOLVING TECHNIQUE AND
PROGRAMMING IN C

Time : 3 hours

Full Marks : 60

Instructions : akubihar.com

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **SEVEN** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question Nos. 1 and 2 are compulsory.

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GROUP—A

1. Choose the correct option/Answer the following (any six) : 2×6=12

(a) While calling the fprintf() function in the format string conversion specifier %s can be used to write a character string in capital letters.

- (i) True
- (ii) False

(b) In a call of printf() function, the format specifier %b can be used to print binary equivalent of an integer.

- (i) True
- (ii) False akubihar.com

(Turn Over)

- (c) If an array is declared as
`int a[4] = { 3, 0, 1, 2 }`
 then the values assigned to a[0] and a[4] will be
- (i) 3, 2
 - (ii) 0, 2
 - (iii) 3, 0
 - (iv) 0, 4 akubihar.com
- (d) The process of repeating a group of statements in an algorithm is known as
- (i) sequence
 - (ii) iteration
 - (iii) flow
 - (iv) selection
- (e) The individual units of a C program are known as
- (i) records
 - (ii) tokens
 - (iii) units
 - (iv) program akubihar.com
- (f) How many values a function can return at a time?
- (i) Only 1
 - (ii) Depends on the system
 - (iii) Infinite values
 - (iv) 2

(g) When compared to call by value, the call by reference is ____ in execution.

(i) fast

(ii) neither slow nor fast

(iii) slow

(iv) equal

(h) Identify the incorrect declaration of arrays from the following :

(i) int a[50]

(ii) float values[10][20]

(iii) double a[50]

(iv) int score[10, 15]

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(i) Which of the following statements is wrong with respect to a storage class?

(i) It specifies the default initial value

(ii) It specifies the life of a variable

(iii) By default a storage class is static

(iv) It specifies where variable is stored

(j) What is 'C' language?

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2. Answer any *three* of the following : $4 \times 3 = 12$

(a) What do you mean by call by value and call by reference? Write a program to swap the two numbers by using third variable with both call by value and call by reference.

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(Turn Over)

(b) What is a storage class? Discuss the various storage classes.

(c) What is an array? What are its merits and demerits? What is 2-dimensional array? How are 2-dimensional arrays initialized? Also explain how array elements are accessed in 2-dimensional arrays.

(d) Write a program to implement a structure variable. Give some differences between structure and union.

(e) Discuss different types of operators using in C language.

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GROUP—C

Answer any *three* questions :

$12 \times 3 = 36$

3. Why are files needed? How we open and close an existing file in C? Also discuss the various modes to open a file.

4. Write a program to print the second largest number between 10 numbers.

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5. Give the advantages of array using pointers. Write a program to access elements of an array by using pointers.

6. Discuss the following :

- (a) Preprocessor commands
- (b) Difference between `calloc()` and `malloc()` functions
- (c) Difference between functions and macros

7. What are the advantages and disadvantages of a recursive function? Write a program to find the sum of first n numbers using recursion.

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Code : 303204

BCA 2nd Semester Theory Examination, 2017

Problem Solving Technique & Programming in C

Time : 3 hours

Full Marks : 60

Instructions :

- (i) The questions are of equal value.
- (ii) There are seven questions in this Paper.
- (ii) Attempt five questions in all.
- (iv) Question No. 1 & 2 is compulsory.

1. Choose the correct answer (any six)

- (i) Which of the following special symbol allowed in a variable name?
 - (a) *(asterisk)
 - (b) | (pipeline)
 - (c) -(hyphen)
 - (d) _(underscore)
- (ii) How would you round off a value from 1.66 to 2.0?
 - (a) cell (1.66)
 - (b) floor (1.66)
 - (c) roundup (1.66)
 - (d) roundto (1.66)

P.T.O.

(iii) Which of the following is not logical operator?

- (a) &
- (b) &&
- (c) | |
- (d) !

(iv) In mathematics and computer programming, which is the correct order of mathematical operators?

- (a) Addition, Subtraction, Multiplication, Division
- (b) Division, Multiplication, Addition, Subtraction
- (c) Multiplication, Addition, Division, Subtraction
- (d) Addition, Division, Modulus, Subtraction

(v) Which of the following cannot be checked in a switch case statement?

- (a) Character
- (b) Integer
- (c) Float
- (d) Enum

(vi) The keyword used to transfer control from a function back to the calling function is

- (a) switch
- (b) go to
- (c) go back
- (d) return

Code : 303204

(vii) What is (void*)0?

- (a) Representation of NULL pointer
- (b) Representation of void pointer
- (c) Error
- (d) None of above

(viii) What is the use of a '\0' character?

(ix) What is the difference between the = symbol and == symbol?

(x) Can I use "int" data type to store the value 32768? Why?

2. Answer any three of the following:

- A. What is macro? Why do we use macro?
- B. What is the difference between while and do-while loops in c?
- C. What is the difference between pre-increment operator and post-increment operator in C?
- D. What is inline function in C?
- E. What is a newline escape sequence?

3. What is scope & storage allocation of register, static and local variables? Explain with Examples.

4. What are storage memory, default value, scope and life of Automatic and Register, Static and External storage class?

5. What are the differences between formal arguments and actual arguments of a function? Explain with an example.

P.T.O.

Code : 303204

3

6. What are the differences between malloc() and calloc()? Explain with Examples.

7. What is an Enumeration? What is a structure? Explain with Examples.

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Code : 303204

BCA 2nd Semester Exam., 2018

**PROBLEM SOLVING TECHNIQUE AND
PROGRAMMING IN C**

Time : 3 hours

Full Marks : 60

Instructions :

- (i) All questions carry equal marks.
- (ii) There are **SEVEN** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question Nos. 1 & 2 are compulsory.

1. Choose the correct answer of the following (any six) :

- (a) In which header file is the NULL macro defined?
 - (i) stdio.h
 - (ii) stddef.h
 - (iii) stdio.h and stddef.h
 - (iv) math.h

(Turn Over)

8AK/504

(2)

- (b) If a variable is a pointer to a structure, then which of the following operators is used to access data members of the structure through the pointer variable?

- (i) .
- (ii) &
- (iii) *
- (iv) ->

- (c) Which is the stage of the following code?

```
#include<stdio.h>  
gets replaced by the contents of the  
file stdio.h
```

- (i) During editing
- (ii) During linking
- (iii) During execution
- (iv) During preprocessing

- (d) How will you free the allocated memory?

- (i) remove(var-name);
- (ii) free(var-name);
- (iii) delete(var-name);
- (iv) dalloc(var-name);

8AK/504

(Continued)

(e) What is the similarity between a structure, union and enumeration?

- (i) All of them let you define new values
- (ii) All of them let you define new data types
- (iii) All of them let you define new pointers
- (iv) All of them let you define new structures

(f) What does the following declaration mean?
`int (*ptr)[10];`

- (i) ptr is array of pointers to 10 integers
- (ii) ptr is a pointer to an array of 10 integers
- (iii) ptr is an array of 10 integers
- (iv) ptr is a pointer to array

(g) When is the 'void' keyword used in a function?

(h) How do you generate random numbers in C?

(i) What are actual arguments?

(j) What are formal parameters?

2. Answer any three of the following :

- (a) What is an endless loop?
- (b) What are comments and how do you insert it in a C program?
- (c) What is variable initialization and why is it important?
- (d) What is pointer to a function?
- (e) What is type casting?

3. What are command line arguments of main function? Explain with a suitable example.

4. What are header files? Are functions declared or defined in header files? <https://www.akubihar.com>

5. What is a pointer in C? What are the advantages of using pointers? Explain with examples.

6. What is a structure? Explain structure within structure with an example.

7. What are the differences between formal arguments and actual arguments of a function? Explain with an example.

★ ★ ★

(Turn Over)

Code : 303204

BCA 2nd Semester Exam., 2019

**PROBLEM SOLVING TECHNIQUE AND
PROGRAMMING IN C**

Time : 3 hours

Full Marks : 60

Instructions :

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **SEVEN** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question Nos. **1 & 2** are compulsory.

**1. Choose the correct answer of the following
(any six) : 2×6=12**

- (a) Define keyword.
- (b) Define structure.
- (c) What is compiler?
- (d) What is the difference between operators and operand?

AK9/775

(Turn Over)

- (e) Define local variable and global variable with examples.
- (f) Define recursion.
- (g) Define union.
- (h) Define pointer.
- (i) What is identifier?

2. Answer any three of the following : 4×3

- (a) Explain the switch-case statement with syntax.
- (b) Explain structure with example.
- (c) What is file? Write its operations.
- (d) Explain numeric data types used in C.
- (e) Explain union with example.

3. Discuss the Dynamic Memory Management of C.

4. Explain the different types of loops in C with syntax and give example.

5. Explain "if" and "switch" with example.

AK9/775

(Continue)

(3)

6. Explain pointer. Give example of passing pointer to function. 12
7. What is array? Explain its types with example. 12

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Code : 303204

AK9—650/775

BCA 2nd Semester Exam., 2021

PROBLEM SOLVING TECHNIQUE
AND PROGRAMMING IN C

Time : 3 hours

Full Marks : 60

Instructions :

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **SEVEN** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question Nos. 1 and 2 are compulsory.

1. Answer/Choose the correct option of the following (any six) : 2×6=12

- (a) perror() function used to
- (i) work same as printf()
 - (ii) print the error message specified by the compiler
 - (iii) print the garbage value assigned by the compiler
 - (iv) None of the above

(b) Which operators are known as ternary operators?

- (i) ::, ?
- (ii) ?, :
- (iii) ?, ::
- (iv) None of the above

(c) What is a pointer?

(d) What are the differences between malloc() and calloc()?

(e) Define recursion.

(f) Differentiate between local variables and global variables.

(g) What is the difference between strings and arrays?

(h) What are the different types of real data type in C?

- (i) float, double
- (ii) short int, double, long int
- (iii) float, double, long double
- (iv) double, long int, float

(i) The keyword used to transfer control from a function back to the calling function is

- (i) switch
- (ii) goto
- (iii) go back
- (iv) return

(j) What is the output of this C code?

```
#include<stdio.h>
int main()
{
    int i=0;
    for(;;)
        printf("In for loop\n");
        printf("After loop\n");
}
```

- (i) Compile time error
- (ii) Infinite loop
- (iii) After loop
- (iv) Undefined behavior

2. Answer any *three* of the following : 4×3=12

- (a) Write down an algorithm to find the largest data value of a set of given data values.
- (b) What are the key features in the C programming language?
- (c) What are the basic data types associated with C?
- (d) What is the description for syntax errors? What is the process to create increment and decrement statements in C? Give example.

(e) Describe wild pointers in C. Describe the difference between = and == symbols in C programming. What is the explanation for prototype function in C?

- 3. What is the difference between call by value and call by reference? Explain with example. Some coders debug their programs by placing comment symbols on some codes instead of deleting it. How does this aid in debugging? 12
- 4. What are header files and what are its uses in C programming? What is syntax error? 12
- 5. What are variables and in what way it is different from constants? Explain with example. What are preprocessor directives? 12
- 6. Differentiate between algorithm and flowchart. 12
- 7. Write a C program to check whether a number is palindrome or not. 12

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BCA 2nd Semester Exam., 2022

PROBLEM-SOLVING TECHNIQUE AND
PROGRAMMING IN C

Time : 3 hours

Full Marks : 60

Instructions :

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **SEVEN** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question Nos. 1 and 2 are compulsory.

1. Answer any six of the following as directed :

2×6=12

- (a) What will be the output of the following statement?
- ```
printf("%X%X%ci%X", 11, 10, 's', 12);
```
- (i) error
  - (ii) basc
  - (iii) Bas94c
  - (iv) None of the above

( Choose the correct option )

- (b) What will be the output of the following statement?

```
long int a = scanf("%ld%ld", &a, &a);
printf("%ld", a);
```

- (i) error
- (ii) garbage value
- (iii) 0
- (iv) 2

( Choose the correct option )

- (c) What will be the output of the following program?

```
#include void main() {
int a = 2; switch (a)
{
case 1:printf("goodbye");
break;
case 2:continue;
case 3:printf("bye");
}
}
```

- (i) error
- (ii) goodbye
- (iii) bye
- (iv) byegoodbye

( Choose the correct option )

- (d) What is file pointer?
- (e) What is debugging?
- (f) How do you generate random numbers in C?
- (g) What is the significance of an algorithm to C programming?
- (h) What is the modulus operator?
- (i) Give the difference between \* and & in C pointer.
- (j) What are command line arguments?

2. Answer any *three* of the following :  $4 \times 3 = 12$

- (a) How are global variables different from static variables?
- (b) Compare static memory allocation with dynamic memory allocation.
- (c) Explain the role of protected access specifier. What is keyword?
- (d) Explain tokens in C.
- (e) Briefly explain the various file opening modes in C.

- 3. (a) Explain the two-way selection (if, if-else, nested if-else, cascaded if-else) in C language with syntax. 8
- (b) Without using if-else statement, write a program in C to calculate the result as follows : 4  
If Marks is less than 50, Result is 'FAIL' and if Marks is greater than or equal to 50, Result is 'PASS'.
- 4. (a) Explain the different types of loops in C with syntax and example. 8
- (b) Show how break and continue statements are used in a C program with example. <https://www.akubihar.com> 4
- 5. What is a pointer? Explain how the pointer variable declared and initialized. Explain the array of pointers with example. 12
- 6. Explain how the structure variable passed as a parameter to a function with example. 12
- 7. (a) What is the difference between sequential and random access files? 4
- (b) Write a C program to reverse the content of the file and print it. 8

\*\*\*

**Code : 303204**

**BCA 2nd Semester Exam., 2025**

**PROBLEM-SOLVING TECHNIQUE AND  
PROGRAMMING IN C**

*Time : 3 hours*

*Full Marks : 60*

*Instructions :*

- (i) The marks are indicated in the right-hand margin.*
- (ii) There are **SEVEN** questions in this paper.*
- (iii) Attempt **FIVE** questions in all.*
- (iv) Question Nos. **1** and **2** are compulsory.*

**1.** Answer/Choose the correct option of the following (any six) : 2×6=12

- (a) What is the role of break statement in C program?*
- (b) What is the use of goto statement in C program?*

**26AK/205**

*( Turn Over )*

(c) Which of the following is true about the following program?

```
#include <iostream>
class Test {
public:
 inti;
 void get();
};
void Test::get(){
 std::cout<<"Enter the value of i:";
 std::cin>>i;
}
Test t;//Global object
int main() {
 Test t;//local object
 t.get();
 std::cout<<"value of i in local t:"<<t.i<<'\\n';
 ::t.get();
 std::cout<<"value of i in global t:"<<::t.i<<'\\n';
 return 0;
}
```

- (i) Compiler error : Cannot have two objects with same class name
- (ii) Compiler error in line "::t.get();"
- (iii) Compiles and runs fine
- (iv) None of the above

(d) Comment on the output of the following code :

```
1. #include <stdio.h>
2. main()
3. {
4. char*p = 0;
5. *p = 'a';
6. printf("value in pointer p is %c\n", *p);
7. }
```

- (i) It will print a
- (ii) It will print 0
- (iii) Compile time error
- (iv) Runtime error

(e) What is the output of this C code?

```
1. #include <stdio.h>
2. main()
3. {
4. if(sizeof(int)>-1)
5. printf("True");
6. else
7. printf("False");
8. }
```

- (i) True
- (ii) False

(f) What is the output of this C code?

```
1. #include <stdio.h>
2. main()
3. {
4. char*p="Sanfoundry C Test";
5. p[0]='a';
6. p[1]='b';
7. printf("%s", p);
8. }
```

(i) abnfoundry C Test

(ii) Sanfoundry C Test

(iii) Compile time error

(iv) Runtime error

(g) Which of the following is a correct format for declaration of function?

(i) return-type  
function-name(argument type);

(ii) return-type  
function-name(argument type) {}

(iii) return-type  
(argument type)function-name;

(iv) Both (i) and (ii)

( 5 )

(h) Which of the following function declarations is illegal?

- (i) `int lbhk(int);`
- (ii) `int lbhk(int a);`
- (iii) `int 2bhk(int*, int []);`
- (iv) All of the above

(i) The value obtained in the function is given back to main by using \_\_\_\_\_ keyword.

- (i) `return`
- (ii) `static`
- (iii) `new`
- (iv) `volatile`

(j) Why and when do we use the `#define` directive and `#include` directive?

2. Answer any *three* of the following : 4×3=12

(a) Distinguish between the following pairs :

- (i) Variables and Identifiers
- (ii) `int main()` and `void main()`

- (b) Write a program in C to find whether the entered character is an alphabet or digit or any other special character.
  - (c) What is a pointer? What are the uses of a pointer?
  - (d) What are structure and union? What are the differences between structure and union?
  - (e) Differentiate between pass-by-reference and pass-by-value. What is static identifier?
3. Explain preprocessor. What are the advantages of preprocessor? What are the facilities provided by preprocessor? 12
4. What are static memory allocation and dynamic memory allocation? What is the purpose of realloc? What is a pointer to a pointer? Explain with example. 12
5. Describe about storage allocation and scope of global, extern, static, local and register variables. What is recursion? What is the difference between strings and arrays? 12



6. What are macros? What are its advantages and disadvantages? Out of fgets() and gets() which function is safe to use and why? 12
7. What is FILE? How is fopen() used? How is a file closed? What is a random access file? What is the purpose of ftell? What is the purpose of rewind()? 12

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